Trustworthy Causal Inference Under Distribution Shifts

Causal Data Generation

Research Progress Note — 26 October 2026

1. Motivation and Research Question

Following our earlier discussion on formulating a concrete scientific problem, I started by constructing a controlled synthetic dataset with a known causal data-generating mechanism.

The objective is to establish a mathematically verifiable benchmark for investigating causal effect estimation, Bayesian uncertainty, explainability, and reliability under distribution shifts.

Research question: Can observational prediction differ from the true effect of an intervention, even when the underlying data-generating mechanism is known?

2. Structural Causal Model

Consider a machine whose workload influences both its operating speed and energy consumption. Operating speed also directly influences energy consumption.

We define:

- U: workload (observed confounder),
- X: operating speed (intervention variable),
- Y: energy consumption (outcome).

The causal graph is $U \to X$, $U \to Y$, and $X \to Y$.

The structural equations are:

$$U \sim \mathcal{N}(\mu_e, \sigma_e^2), \tag{1}$$

$$X = 0.8U + \epsilon_X, \tag{2}$$

$$Y = 2X + 1.5U + 0.5XU + \epsilon_Y, \tag{3}$$

where $\epsilon_X, \epsilon_Y \sim \mathcal{N}(0, 1)$ are mutually independent and independent of U.

The interaction term $0.5XU$ introduces heterogeneous treatment effects, while U creates the confounding path $X \leftarrow U \to Y$.

3. Ground-Truth Causal Effects

Under an intervention $do(X = x)$, the treatment assignment equation is replaced by $X = x$.

The corresponding potential outcome is,

$$Y(x) = 2x + 1.5U + 0.5xU + \epsilon_Y. \tag{4}$$

Hence, the individual treatment effect is,

$$Y(1) - Y(0) = 2 + 0.5U. \tag{5}$$

The population average treatment effect (ATE) is,

$$\text{ATE}_e = \mathbb{E}[Y(1) - Y(0)] = 2 + 0.5\mu_e. \tag{6}$$

For environments with $\mu_e = 0$ and $\mu_e = 2$, the theoretical ATE values are 2 and 3, respectively.

4. Identification and Verification

Because workload U blocks the backdoor path, the interventional distribution is identifiable through adjustment,

$$p(y \mid do(X = x)) = \int p(y \mid X = x, U = u)p(u)\, du. \tag{7}$$

The construction satisfies consistency, conditional exchangeability given U, and theoretical positivity for finite treatment values.

For the first environment,

$$\mathbb{E}[Y \mid do(X = x)] = 2x, \tag{8}$$

$$\mathbb{E}[Y \mid X = x] = \frac{112}{41}x + \frac{10}{41}x^2. \tag{9}$$

The difference reveals how observational association can differ from causal intervention response.

I've put everything into a Jupyter notebook, where I generated the dataset, worked out the analytical treatment effects, and ran numerical checks to make sure the results match our equations.

5. Experimental Scope and Next Steps

I've generated the training, validation, and test datasets with 500, 2,000, and 10,000 training samples across two workload environments.

So far, I've established the causal ground truth, but I haven't tested the machine learning models yet.

Next, I'll compare ordinary regression, confounder-adjusted regression, and nonlinear machine learning models using both prediction error and causal-effect estimation error, and share the results with you.

After that, I'll move on to Bayesian inference, latent confounding, causal representations, and distribution shifts.